1. Construct a comprehensive concept map explaining the foundations of Network Security. Connect network security fundamentals with security objectives, the CIA triad (Confidentiality, Integrity and Availability), network assets, vulnerabilities, threats, and protective measures and explain how these concepts collectively support secure network operations.

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98/100 Good

Network Security Fundamentals
- Protects data, systems and communications within a network.
- Uses policies, procedures, technologies and people.
- Ensures reliable and secure network operations.

Can be exploited by / Supports

Security Objectives
- Protect data and resources.
- Prevent unauthorized access
- Minimize damage from attacks.

achieved through

The CIA Triad
- Confidentiality → Only authorized users can access the data.
- Integrity → Data remains accurate and unchanged
- Availability → Resources are accessible when needed.

guides the protection

Network Assets
- Hardware - Routers, Switches, firewalls, servers
- Software - OS, applications
- Data - Files, databases, credentials.
- Network infrastructure - cables, protocols, bandwidth

can be exploited by

Vulnerabilities
- Unpatched software/ outdated systems.
- Weak password/ poor authentication
- Misconfigurations
- Human errors
- Insecure protocols/ lack of encryption

are used by

Threats
=> Malware (virus, worms).
=> Hackers / cybercriminals.
=> Insider threats (malicious or accidential).
=> Phishing and social engineering.
=> Denial of service
=> Man-in-the-Middle attacks.

are countered by

Protective Measures
-> Firewalls (filter traffic).
-> Encryption (protects data)
-> Authentication & Access control.
-> IDS/IPS (detects & blocks)
-> Regular updates and patch managements.
-> Backups & disaster recovery.

work together to maintain

Secure Network Operations
- Confidential data remains
- Data and systems stay accurate
- Services and network remain available
- Risks and attack impacts are minimized
- Builds trust & enables safe communication.

When fundamentals, objectives, assets and controls work together, the network stays secure, resilient and trustworthy

2. Develop a concept map illustrating the CIA Triad as the foundation of network security. Relate Confidentiality, Integrity and Availability to different types of network threats and demonstrate how a single threat can affect one or more security goals.

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The CIA Triad - Foundation of Network Security

The CIA Triad represents the core goals of network security. Together, they ensure that data and resources are protected.

Confidentiality

- Protects sensitive information
- Ensures only authorized users can access the network and its resources.
- Achieved through encryption, access control, authentication, etc.

Integrity

* Prevents unauthorized modification of data or system settings.
* Maintains trust and reliability of information.
* Achieved through hashing, digital signatures, checksums.

Availability

* Prevents disruption or denial of services
* keeps the network, applications and data available to authorized users
* Achieved through redundancy, backups, etc.

Types of Network Threats

Malware

*Steals data (C).
*Alters/Deletes file (I).
*Disrupts services (A).

Phishing / Social Engineering

=> Steals credentials (C)
=> Gives unauthorized access (I)
=> Can lead to service disruption (A)

Denial of Service

→Can expose system information (C)
→Can cause data corruption (I)
→Makes service unavailable (A)

Man-in-the Middle

*Intercepts data (C)
*Can modify data (I).
*May block / slow communication (A)

Insides threat

oLeaks confidential data (C)
oModifies or manipulates data (I)
oCan sabotage or remove services (A).

Physical Threats

*Exposes data (C)
*Tampering with devices (I).
*Causes downtime (A)